chapter. The frequency response can be plotted on a linear vertical axis, such as in (c), or on a logarithmic scale (decibels), as shown in (d). The linear scale is best at showing the passband ripple and roll-off, while the decibel scale is needed to show the stopband attenuation.

Don't remember decibels? Here is a quick review. A **bel** (in honor of Alexander Graham Bell) means that the power is changed by a *factor of ten*. For example, an electronic circuit that has 3 bels of amplification produces an output signal with $10 \times 10 \times 10 = 1000$ times the power of the input. A **decibel (dB)** is one-tenth of a bel. Therefore, the decibel values of: -20dB, -10dB, 0dB, 10dB & 20dB, mean the power ratios: 0.01, 0.1, 1, 10, & 100, respectively. In other words, every *ten* decibels mean that the power has changed by a factor of ten.

Here's the catch: you usually want to work with a signal's *amplitude*, not its *power*. For example, imagine an amplifier with 20dB of gain. By definition, this means that the power in the signal has increased by a factor of 100. Since amplitude is proportional to the square-root of power, the amplitude of the output is 10 times the amplitude of the input. While 20dB means a factor of 100 in power, it only means a factor of 10 in amplitude. Every *twenty* decibels mean that the amplitude has changed by a factor of ten. In equation form:

EQUATION 14-1
Definition of decibels. Decibels are a way of expressing a *ratio* between two signals. Ratios of power (P_1 & P_2) use a different equation from ratios of amplitude (A_1 & A_2).

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

$$\text{dB} = 20 \log_{10} \frac{A_2}{A_1}$$

The above equations use the base 10 logarithm; however, many computer languages only provide a function for the base *e* logarithm (the natural log, written $\log_e x$ or $\ln x$). The natural log can be use by modifying the above equations: $\text{dB} = 4.342945 \log_e(P_2/P_1)$ and $\text{dB} = 8.685890 \log_e(A_2/A_1)$.

Since decibels are a way of expressing the ratio between two signals, they are ideal for describing the gain of a system, i.e., the ratio between the output and the input signal. However, engineers also use decibels to specify the amplitude (or power) of a *single* signal, by referencing it to some standard. For example, the term: **dBV** means that the signal is being referenced to a 1 volt rms signal. Likewise, **dBm** indicates a reference signal producing 1 mW into a 600 ohms load (about 0.78 volts rms).

If you understand nothing else about decibels, remember two things: First, -3dB means that the amplitude is reduced to 0.707 (and the power is